

AI-DRIVEN MAINTENANCE

# Anomaly Detection & Fatigue Intelligence for Aerospace and Rail Assets

A LangGraph multi-agent digital twin where physics and ML negotiate, a Gemini Judge investigate with multiple tools, and every decision leaves an audit trail.



## THE CHALLENGE

# Fleet failures are expensive and mostly unavoidable

Aircraft and rail operators still rely on fixed inspection intervals and manual thresholds. By the time a gauge trips, damage has often already accumulated - turning a routine swap into an unscheduled grounding.



### Reactive, not predictive

Threshold alarms fire only after a fault is already underway.



### Unplanned downtime

Emergency repairs cost 3-5x more than scheduled maintenance.



### Fragmented data

Sensor CSVs, material specs, and fatigue models live in silos.

## INDUSTRY BENCHMARK

Unplanned maintenance costs the aerospace industry billions annually - and a single grounded aircraft can idle a fleet's economics for days.

**3-5x**

cost of reactive vs. planned repair

**6-9 mo**

typical AI-maintenance rollout

**3-5x**

projected first-year ROI

ASSET COVERAGE

# Seven components, purpose-built sensor profiles

Each profile defines normal / warning / critical bands per sensor, tuned separately for aircraft and rail duty cycles.



Engine

Vibration · Temp · Oil Pressure · Oil Temp

INCONEL718



Wing

Vibration · Load Factor · Temperature

AL7075-T6



Fuselage

Vibration · Temperature · Load Factor

AL7075-T6



Landing Gear

Vibration · Load Factor · Pressure

STEEL4340



Brake

Temperature · Pressure · Acoustic Emission

STEEL4340



Bogie

Vibration · Load Factor · Temperature

CASTIRON



Wheel

Vibration · Temperature · Pressure

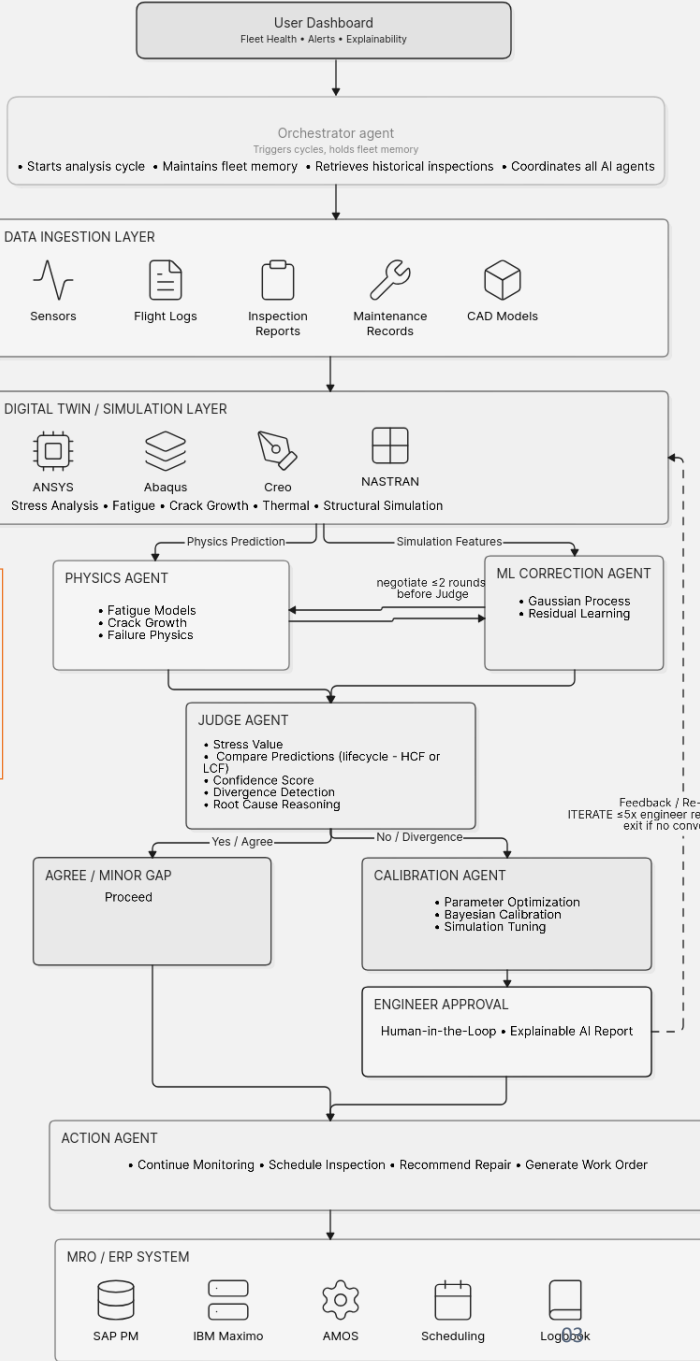
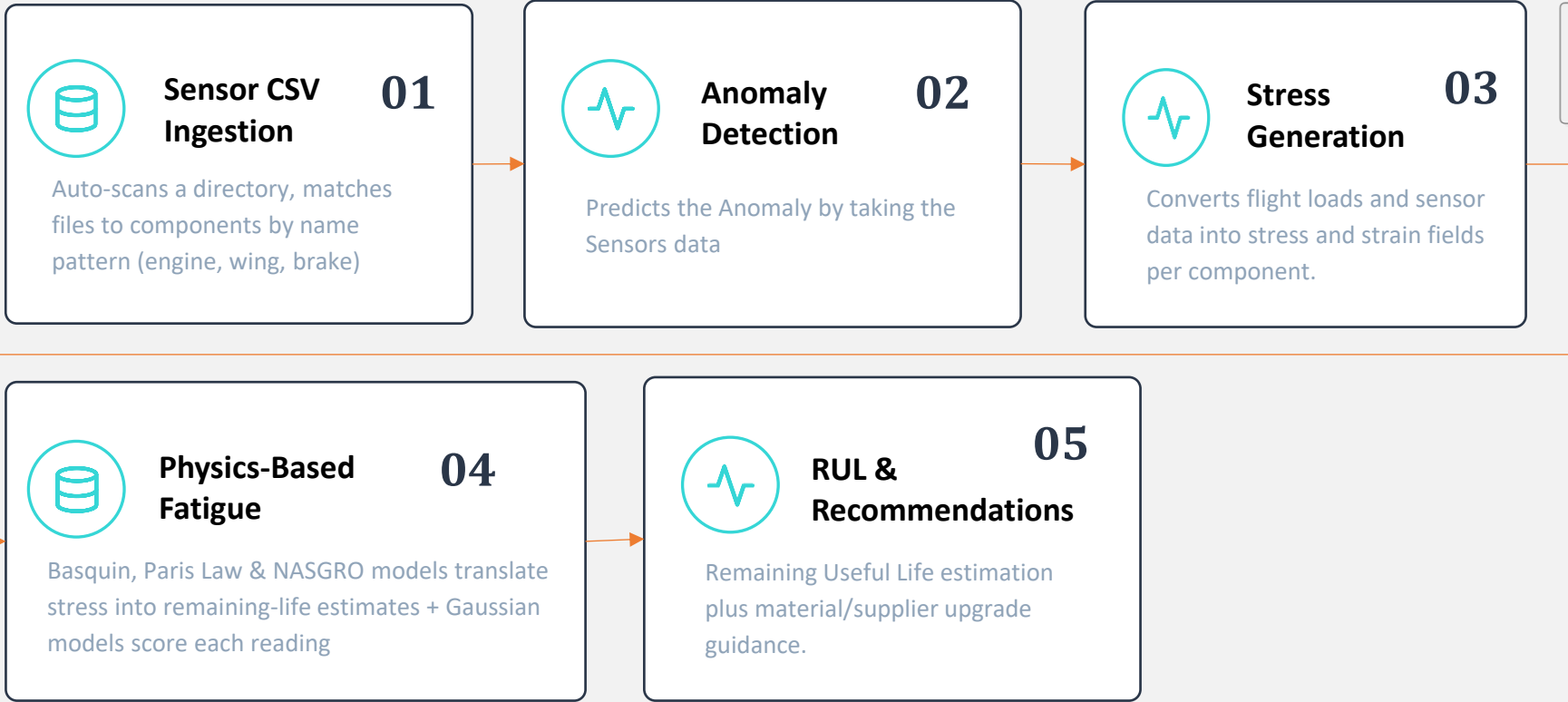
STEEL4340

Component profiles auto-map to CSV columns by name pattern no manual schema wiring required.

SYSTEM ARCHITECTURE

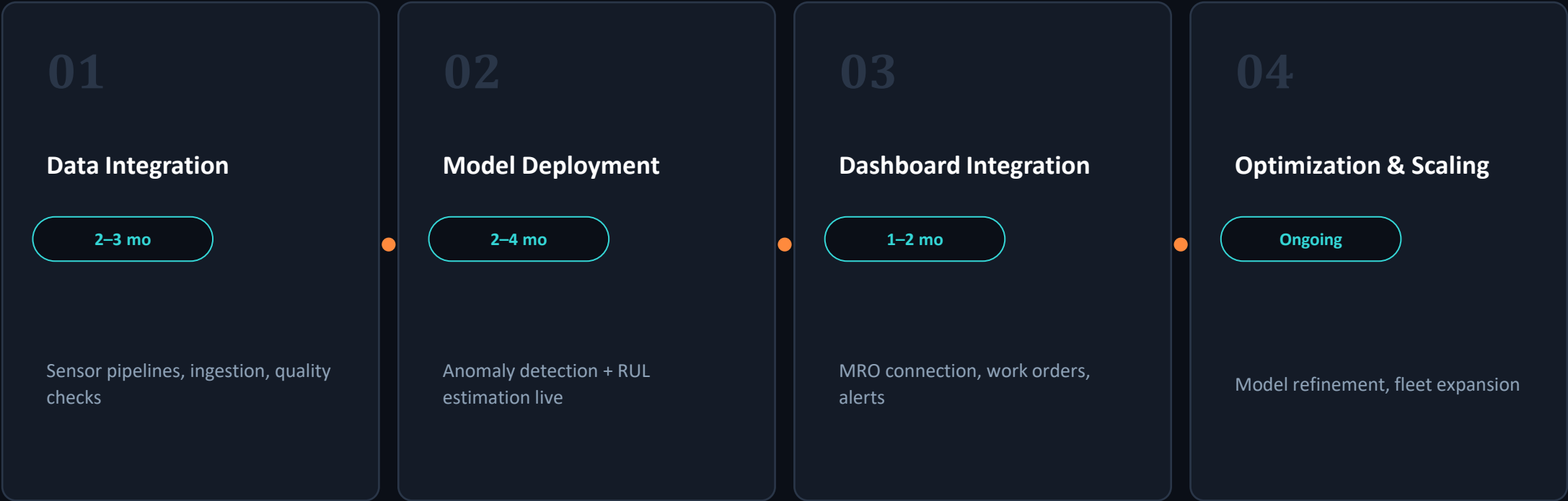
# Two engines, one fleet-health pipeline

A lightweight CSV-threshold detector for rapid field triage, feeding into a deeper AI + physics engine for lifecycle-grade decisions.



# From data pipeline to fleet-wide rollout

A phased integration path connects the model directly into existing MRO and asset-management workflows.



TIMELINE

**6–9 months**

INVESTMENT

**\$250K–\$500K**

PROJECTED ROI

**3–5x, Year 1**

# Every anomaly ends in a decision, not just a flag

The material database cross-references current alloys against a curated upgrade map — surfacing supplier, cost, and confidence in one pass.

| Current    | Recommended     | Key Gain          | Risk   | Conf. |
|------------|-----------------|-------------------|--------|-------|
| Inconel718 | Inconel718Plus  | +25% fatigue life | Low    | 82%   |
| Steel4340  | Steel300M       | +20% fatigue life | Medium | 78%   |
| Al7075-T6  | Al7050-T7451    | +15% toughness    | Low    | 80%   |
| CastIron   | DuctileCastIron | +25% fatigue life | Low    | 83%   |



Recommendations bundle estimated cost, supplier lead time, and projected savings — so engineering and procurement work from the same number.

# Turn every sensor reading into a maintenance decision

Maintenance System · Anomaly Detection & Fatigue Intelligence Engine

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Questions & next steps welcome.